Examiner only

11. (a) Arsenic oxide, As_2O_3, is prepared on an industrial scale by roasting arsenic-containing ores such as arsenopyrite, FeAsS, in air. The other products formed are iron(III) oxide and sulfur dioxide.

(i) State the oxidation state of arsenic in As_2O_3. [1]

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(ii) Give a balanced chemical equation for the industrial production of As_2O_3 from FeAsS. [2]

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(b) As_2O_3 is moderately soluble in water. 100 cm^3 of a saturated solution at 25 °C contains 2.06 g.

When dissolved in water, the oxide reacts to form arsenous acid.

$$As_2O_3 + 3H_2O \longrightarrow 2H_3AsO_3$$

(i) Calculate the concentration of the arsenous acid, in mol dm^{-3}, in the saturated solution. [3]

concentration of H_3AsO_3 = ... mol dm^{-3}

(ii) A solution of arsenous acid has a pH of 5.11.

Calculate the hydrogen ion concentration of this solution. [1]

$[H^+]$ = ... mol dm^{-3}

(c) The formula for arsenous acid can be written as $As(OH)_3$ since it contains three hydroxyl (OH) groups bonded to arsenic.

Suggest the shape around the arsenic atom in $As(OH)_3$. Justify your answer by using VSEPR theory. [3]

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(d) Phosphorus can form two chlorides, PCl_3 and PCl_5.

0.181 g of a chloride of phosphorus gave 39 cm^3 of vapour at 1 atm pressure when heated to 87 °C. The sample was completely vapourised.

Show that the chloride of phosphorus was PCl_3. [4]

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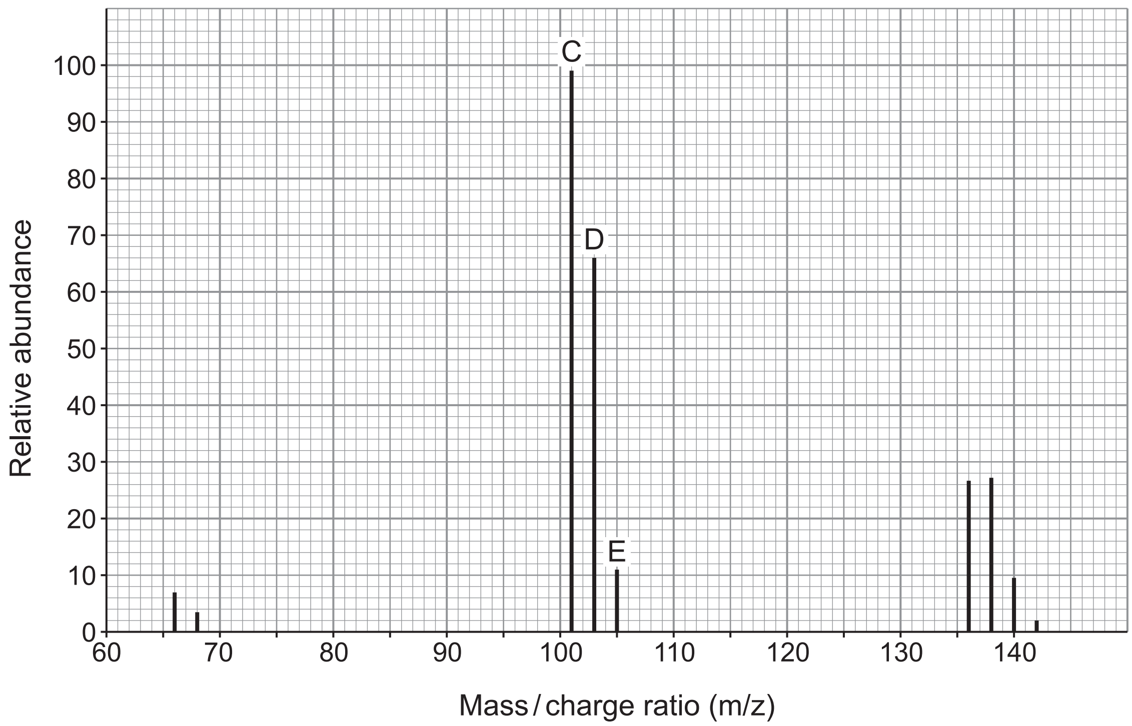
(e) The molecular ion region of the mass spectrum of PCl_3, is shown below.



(i) Identify the species responsible for peak C at m/z 101. [1]

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(ii) Explain why the height ratio of peaks C:E is 9:1. [2]

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